HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND
COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS
AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

10 December 2004

Time allowed 2 hours

Attempt all questions

Total Marks 60

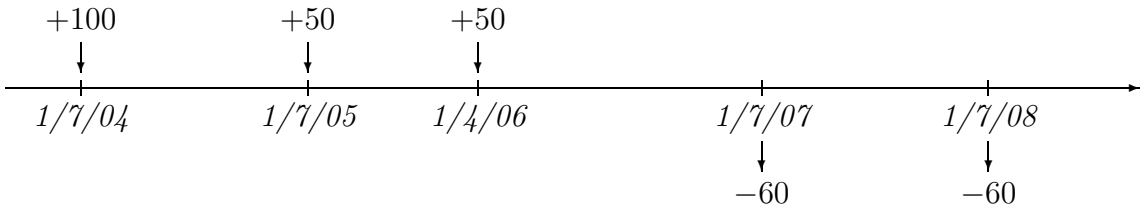
Only approved calculators may be used.

1. Fill in the missing values in the following table:

i	v	d	$i^{(3)}$	$d^{(2)}$
?	0.95	?	?	?

[4 marks]

2. The effective rate of interest will be 4% per annum during 2004 and 2005, and 2% per half-year thereafter. A series of cashflows is given as follows:



   (a) What is the present value on 1 July 2004 of these cashflows?

   (b) What is the accumulation on 1 January 2009 of these cashflows?

[7 marks]

3. Evaluate the following:

   (a) $s_{\overline{17}|}^{0.11}$.

   (b) $\ddot{a}_{\overline{\infty}|}^{0.065}$.

   (c) ${}_{25}|a_{\overline{40}|}^{0.06}$.

[6 marks]

continued ...

4. (a) Prove from first principles that:

$$(Is)^{i}_{\overline{n}|} = \frac{\ddot{s}^{i}_{\overline{n}|} - n}{i}.$$

[3 marks]

(b) An annuity-certain is payable annually in arrear for 20 years. The first payment is £5,000 and subsequent payments increase by £200 per year. What is the present value of this annuity at outset:

(i) If the effective rate of interest is 10% per annum throughout?

(ii) If the effective rate of interest is 10% per annum during the first 19 years and 8% per annum during the last year?

[6 marks]

[Total 9 marks]

5. (a) You borrow £10,000 to buy a car. The loan is to be repaid over 3 years by level monthly repayments, payable in arrear, and the interest rate is 12.682503% per annum effective.

(i) What is the level monthly repayment?

(ii) What is the capital outstanding just after the 12th repayment has been made?

[4 marks]

(b) Just after the 12th repayment has been made, you ask the lender if you can immediately repay £2,000 of the capital outstanding. The lender agrees.

(i) What is the new monthly repayment?

(ii) What is the capital content of the 13th repayment?

[4 marks]

[Total 8 marks]

continued ...

6. What is the price per £100 nominal, of a bond with remaining term 14 years, with coupons of 6% per annum payable half-yearly in arrear, redeemable at par, if the yield obtained by an investor subject to income tax at 20% and capital gains tax at 30% is 9% per annum effective?

[7 marks]

7. An investment fund, which has assets worth £10,000 on 31 December 2002, receives inflows of £1,000 on 1 January in 2003 and 2004, and an outflow of £5,000 on 1 January 2005. On 31 December 2005 it has assets of £8,586.

   (a) Show that the money-weighted rate of return during the years 2003–2005 is 5% per annum.

   [1 mark]

   (b) Suppose that the fund has assets of £13,420 on 31 December 2004. Show that the linked internal rate of return during the years 2003–2005 is 4.64% per annum, based on the subdivision of the period into 2003–2004 and 2005.

   [4 marks]

   (c) Suppose also that the fund has assets of £12,000 on 31 December 2003. Show that the time weighted rate of return during the years 2003–2005 is 4.72% per annum.

   [2 marks]

[Total 7 marks]

8. You have a sum of money to invest, and you have to choose between investments A, B or C, described below. You decide that you will buy the investment with the highest yield. Which do you choose?

   (A) A bond with term 10 years, and coupons of 10% per annum payable in arrear, about to be issued at par and also redeemable at par.

   (B) A bond with term 20 years, and coupons of 8% per annum payable half-yearly in arrear, currently priced at £75 per £100 nominal, and redeemable at par.

   (C) An irredeemable bond (*i.e.* a perpetuity) with coupons payable yearly in arrear. The next coupon, payable in 1 year's time, is £10.50, and thereafter coupons increase at 5% per annum compound. The price of the bond is £150.

[12 marks]

Total 60 marks